# CY1010 – Environmental Chemistry

## Practice Question Paper 3 – Advanced (50 marks)

*Compiled from course slides — not an official department paper. Questions here require calculation, multi-step reasoning, or applying a concept to a new scenario, rather than recalling an isolated fact.*

**Student name: ________________________________________________**

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*Each question carries 1 mark. Choose the single best answer and write the corresponding letter on the "Ans" line. Rough work may be needed for calculation-based questions.*

**1.** In the fission reaction shown in the slides, $^{235}U + n \rightarrow {}^{236}U \rightarrow {}^{144}Ba + {}^{89}Kr + x \cdot n$ + energy, conservation of mass number requires that x equals

a) 1
b) 2
c) 3
d) 4
Ans: ______

**2.** Using the Sun's total power output ($3.84 \times 10^{26}$ W) and the Earth–Sun distance ($1.49 \times 10^{11}$ m) given in the slides, the intensity of solar radiation at Earth's orbit ($I = P / 4\pi r^2$) works out closest to

a) 137.7 W/m²
b) 1377 W/m²
c) 13,770 W/m²
d) 377 W/m²
Ans: ______

**3.** Using Earth's radius ($6.371 \times 10^6$ m) to find the cross-sectional area of Earth's disc, and a solar intensity of 1377 W/m² at Earth's orbit, the total power intercepted by Earth is closest to

a) $1.27 \times 10^{14}$ W
b) $1.755 \times 10^{17}$ W
c) $1.755 \times 10^{20}$ W
d) $1.27 \times 10^{17}$ W
Ans: ______

**4.** The slides estimate that producing 500 EJ of energy via $C + O_2 \rightarrow CO_2$ ($\Delta H \approx -400$ kJ/mol) generates about $1.25 \times 10^{15}$ moles of $CO_2$ per year. Using the same method, how many moles of $CO_2$ would correspond to generating 100 EJ of energy?

a) $6.25 \times 10^{13}$
b) $1.25 \times 10^{14}$
c) $2.5 \times 10^{14}$
d) $2.5 \times 10^{15}$
Ans: ______

**5.** Using the slides' estimate of ~$1.25 \times 10^{15}$ moles of $CO_2$ produced globally per year (from the 500 EJ energy-demand calculation) and a molar mass of $CO_2$ of 44 g/mol, the corresponding mass of $CO_2$ is closest to

a) $5.5 \times 10^{10}$ metric tons
b) $5.5 \times 10^{7}$ metric tons
c) $5.5 \times 10^{13}$ metric tons
d) $5.5 \times 10^{16}$ metric tons
Ans: ______

**6.** The slides state the available wind energy resource is about $1.26 \times 10^9$ MW. Converting this to electricity at 10% efficiency gives approximately

a) $1.26 \times 10^6$ MW
b) $1.26 \times 10^7$ MW
c) $1.26 \times 10^8$ MW
d) $1.26 \times 10^{10}$ MW
Ans: ______

**7.** According to the slides, coal reserves would last about 200 years at current usage, but only about 65 more years if usage grows at 2% per year. Approximately by what factor does this growth in usage shorten the reserve lifetime?

a) About 1.5×
b) About 2×
c) About 3×
d) About 10×
Ans: ______

**8.** Anthracite (90% C, 10% ash), bituminous (80% C, 20% ash), and lignite (70% C, 30% ash) are burned in equal masses. Based on carbon content alone, which produces the greatest mass of $CO_2$ per kg of coal burned?

a) Anthracite
b) Bituminous
c) Lignite
d) All produce equal $CO_2$
Ans: ______

**9.** A hydrogen plant uses steam methane reforming ($CH_4 + H_2O \rightarrow CO + 3H_2$) but installs equipment to capture and permanently store all of the resulting $CO_2$. According to the classification scheme in the slides, the hydrogen from this plant should now be labelled

a) Grey hydrogen
b) Black/brown hydrogen
c) Blue hydrogen
d) Green hydrogen
Ans: ______

**10.** In the Ostwald process ($4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$, then $2NO + O_2 \rightarrow 2NO_2$), starting from 4 mol of $NH_3$ and assuming complete conversion at each step, the total moles of $O_2$ consumed across these first two reactions is

a) 5 mol
b) 6 mol
c) 7 mol
d) 9 mol
Ans: ______

**11.** Which of the following correctly orders the steps of solvent-based direct air capture, as described in the slides?

a) $CaCO_3$ is heated to ~900 °C → $CO_2$ reacts with NaOH/KOH → anionic exchange with $Ca(OH)_2$ → CaO reacts with water
b) $CO_2$ reacts with NaOH/KOH → anionic exchange with $Ca(OH)_2$ forms $CaCO_3$ → $CaCO_3$ is heated (~900 °C) to release $CO_2$ and CaO → CaO reacts with water to regenerate $Ca(OH)_2$
c) CaO reacts with water → $CO_2$ reacts with NaOH/KOH → $CaCO_3$ forms → heating regenerates NaOH
d) $Ca(OH)_2$ is heated directly to release $CO_2$ without forming $CaCO_3$
Ans: ______

**12.** A gas adsorbing onto a solid surface has a measured heat of adsorption of 15 kJ/mol, and increasing the temperature reduces the amount adsorbed. Based on the ranges given in the slides, this is best classified as

a) Chemisorption, because it is surface specific
b) Physisorption, because ΔHads falls in the 5–50 kJ/mol range typical of weak, non-activated adsorption
c) Chemisorption, because ΔHads exceeds 50 kJ/mol
d) Neither, since 15 kJ/mol is too low for any adsorption
Ans: ______

**13.** A metal–organic framework is reported to have an average pore diameter of 1.2 nm. According to the IUPAC classification given in the slides, this pore would be classified as

a) Macropore
b) Mesopore
c) Micropore
d) Nanopore
Ans: ______

**14.** Based on the GWP values given in the slides ($CO_2$ = 1, $CH_4$ = 25), releasing 500 g of methane is estimated to have roughly the same 100-year warming impact as releasing how much $CO_2$?

a) 20 g
b) 500 g
c) 2,500 g
d) 12,500 g
Ans: ______

**15.** Using the same logic and the slides' GWP value for nitrous oxide (298), how many grams of $N_2O$ would need to be released to have roughly the same warming impact as releasing 29.8 kg of $CO_2$?

a) 10 g
b) 100 g
c) 1,000 g
d) 29,800 g
Ans: ______

**16.** Which property of methane, contrasted with coal's complex polymeric structure, best explains why burning natural gas produces far less particulate (soot) matter than burning coal?

a) Methane is a gas at room temperature
b) Methane contains only a single carbon atom per molecule
c) Methane has a higher calorific value
d) Methane does not require an odorant
Ans: ______

**17.** In the reaction PhC(=O)Me + $CH_2$=$PPh_3$ $\rightarrow$ PhC(=$CH_2$)Me + O=$PPh_3$ shown in the slides (molecular weights 120, 276, 118, and 278 g/mol respectively), treating the alkene as the desired product, the atom economy is closest to

a) 12%
b) 30%
c) 70%
d) 89%
Ans: ______

**18.** In the hydroformylation example in the slides (alkene + CO + $H_2$ $\rightarrow$ linear aldehyde + branched aldehyde, over a catalyst), the atom economy is stated to be 100%. What is the best explanation for this?

a) Only the linear aldehyde is formed, with no side products
b) Both the linear and branched aldehyde products are considered useful, so none of the reactant atoms end up in a wasted byproduct
c) The reaction produces no products at all
d) CO and $H_2$ do not contribute any atoms to the products
Ans: ______

**19.** According to the case study in the slides, the main advantage of the BHC Company's three-step ibuprofen synthesis over the original Boots six-step process is that

a) It uses more hazardous solvents but is faster
b) It incorporates most of the reactant atoms into the desired product, producing far less waste
c) It eliminates the need for any catalyst
d) It produces ibuprofen with a different chemical structure

Ans: ______

**20.** A biodiesel is produced entirely from genetically engineered algae designed to absorb more $CO_2$ than ordinary plants during growth. Based on the generational classification in the slides, this would be classified as

a) First-generation biofuel
b) Second-generation biofuel
c) Third-generation biofuel
d) Fourth-generation biofuel
Ans: ______

**21.** In the heterogeneous catalysis mechanism given in the slides (adsorption → bond-breaking → migration → new bond formation → desorption), why is the final desorption step essential for the catalyst to keep functioning?

a) It provides the activation energy for the reaction
b) It frees up active sites on the catalyst surface for new reactant molecules
c) It converts the catalyst into a different chemical species
d) It increases the concentration of reactants in solution
Ans: ______

**22.** In a catalytic converter, unburned hydrocarbons and CO are oxidised to $CO_2$ and $H_2O$, while NO and $NO_2$ are simultaneously reduced to $N_2$ and $O_2$. What overall type of chemical transformation is being carried out?

a) Only oxidation reactions
b) Only reduction reactions
c) Simultaneous oxidation (of CO/hydrocarbons) and reduction (of NOx)
d) Only acid-base neutralization
Ans: ______

**23.** Bioremediation of an oil spill is generally slower than in-situ burning but is described in the slides as more environmentally desirable. What is the main disadvantage of in-situ burning that biological remediation avoids?

a) In-situ burning does not remove any oil at all
b) In-situ burning produces smoke and air pollutants
c) In-situ burning is far more expensive than biological methods
d) In-situ burning can only be used on land, not water
Ans: ______

**24.** In β-oxidation, the carboxyl carbon of a fatty acid is designated C-1, and oxidative attack occurs at C-3. Based on this numbering, how many carbons away from C-1 does the key oxidation step take place?

a) Zero (at C-1 itself)
b) One carbon away (C-2)
c) Two carbons away (C-3)
d) Three carbons away (C-4)
Ans: ______

**25.** Both linear and cyclic/aromatic hydrocarbons in detergents and petroleum are oxidised during microbial degradation with the help of cytochrome P-450. What general role is cytochrome P-450 playing in this context?

a) Hydrolysis of ester bonds
b) Activation of molecular oxygen for oxidation of the hydrocarbon
c) Reduction of carboxylic acids to alcohols
d) Dehydration of alcohols
Ans: ______

**26.** Naphthalene (two fused benzene rings, $C_{10}H_8$) is oxidised by $KMnO_4$ under acidic conditions to phthalic acid ($C_8H_6O_4$), as shown in the slides. What has structurally happened to the naphthalene?

a) Both rings are destroyed completely
b) One ring is opened and lost as $CO_2/H_2O$ while the other ring is retained and functionalised with two carboxylic acid groups
c) The two rings are fused into a single larger ring

d) No structural change occurs; only hydrogens are replaced
Ans: ______

**27.** Methanol is metabolised in the body first to formaldehyde, which the slides identify as responsible for neuronal damage. Based on this, why might the toxic effects of methanol poisoning be delayed rather than immediate?

a) Methanol itself is directly toxic and acts instantly
b) The toxic effect depends on methanol first being metabolised into formaldehyde, which takes time
c) Methanol has no toxic effects at all
d) Formaldehyde is produced before methanol enters the body
Ans: ______

**28.** A factory currently disposes of a chemical byproduct in a landfill. Applying the Pollution Prevention Act hierarchy (prevent/reduce → recycle → treat → dispose) from the slides, which would be the SECOND-most preferred alternative, after redesigning the process to prevent/reduce the byproduct at the source?

a) Continue landfill disposal
b) Treat the byproduct to reduce hazard before disposal
c) Find a way to recycle the byproduct
d) Incinerate the byproduct instead
Ans: ______

**29.** A chemical manufacturer redesigns a process to use a catalytic (reusable) amount of an acid instead of a stoichiometric (one-time-use) amount. This change most directly reflects which of the 12 Principles of Green Chemistry?

a) Prevention
b) Catalysis
c) Designing Safer Chemicals
d) Real-time Analysis for Pollution Prevention
Ans: ______

**30.** A company switches its industrial solvent from a petroleum-derived solvent to one derived from corn starch, while keeping the reaction chemistry otherwise identical. This best reflects which of the 12 Principles of Green Chemistry?

a) Use of Renewable Feedstocks
b) Design for Degradation
c) Reduce Derivatives
d) Atom Economy
Ans: ______

**31.** A chemical is classified as capable of causing severe skin burns and irreversible eye damage on contact, but is not flammable, explosive, or acutely toxic. Which GHS pictogram would most directly apply, based on the categories given in the slides?

a) Flame
b) Skull and crossbones
c) Corrosion
d) Exploding bomb
Ans: ______

**32.** According to the GHS system described in the slides, a chemical hazard label uses the signal word "Danger" rather than "Warning". What does this indicate about the hazard's severity?

a) The hazard is less severe than one labelled "Warning"
b) The hazard is more severe than one labelled "Warning"
c) "Danger" and "Warning" indicate identical severity
d) The signal word only applies to environmental hazards
Ans: ______

**33.** A reaction has a percentage yield of 95% but an atom economy of only 20%. Based on the definitions given in the slides, which statement correctly describes this process?

a) The reaction is highly efficient overall, since both figures are interchangeable

b) Almost all the theoretically expected product is obtained, but most of the mass of the reactants ends up in byproducts/waste rather than the desired product
c) An atom economy of 20% means the reaction produces almost no product at all
d) A high yield always implies a high atom economy
Ans: ______

**34.** Given that the troposphere's volume is about $6.13 \times 10^{18}$ m³ (which holds ~80% of the atmosphere's mass) and $CO_2$ makes up about 0.04% of atmospheric volume, the volume of $CO_2$ within the troposphere is closest to

a) $2.45 \times 10^{12}$ m³
b) $2.45 \times 10^{15}$ m³
c) $2.45 \times 10^{18}$ m³
d) $6.13 \times 10^{15}$ m³
Ans: ______

**35.** Ocean water contains roughly 33–37 g of dissolved salt per litre, while drinking water must contain less than 300 mg/L. Approximately how many times more concentrated in dissolved salt is ocean water compared to the upper limit for drinking water?

a) About 10 times
b) About 100 times
c) About 1000 times
d) About 10,000 times
Ans: ______

**36.** Biogas typically contains 55–65% methane with the remainder mostly $CO_2$, whereas natural gas is predominantly methane. Based on this difference, what would you expect regarding the energy content per unit volume of biogas compared to natural gas at the same pressure?

a) Biogas would have a higher energy content because of its $CO_2$ content
b) Biogas would have a lower energy content per unit volume, since a smaller fraction of it is combustible methane
c) Both would have identical energy content per unit volume
d) Energy content per unit volume cannot depend on gas composition
Ans: ______

**37.** The slides state that seawater contains far more uranium than land-based ore deposits, and that this uranium is continuously replenished by steady-state reactions between rocks and water. Why is seawater uranium extraction considered "renewable" despite uranium being a finite element?

a) Because uranium itself is renewable, unlike other elements
b) Because removed uranium is continuously replaced through ongoing natural geochemical processes, unlike a static ore deposit
c) Because seawater uranium does not undergo fission
d) Because uranium concentration in seawater is decreasing over time
Ans: ______

**38.** Based on the ordering implied throughout the slides (e.g., the PPA hierarchy placing prevention/reduction before recycling), which of the "3 Rs" is generally the most preferable from a resource-conservation standpoint?

a) Reduce, because it avoids generating waste or demand in the first place
b) Reuse, because it always requires more energy than recycling
c) Recycle, because it is listed last
d) All three are exactly equally preferable
Ans: ______

**39.** DAC removes $CO_2$ that is diluted in ambient air (~420 ppm), while point-source CCS removes $CO_2$ from concentrated exhaust streams. Based on this difference in starting concentration, which method would you expect to require more energy per tonne of $CO_2$ captured, all else being equal?

a) DAC, because it must extract $CO_2$ from a much more dilute source
b) CCS, because power plant exhaust has no $CO_2$ in it
c) Both require identical energy regardless of concentration

d) Neither requires significant energy
Ans: ______

**40.** The slides project that unmitigated emissions would cause global temperature rise to exceed 4 °C by 2100, while the Paris Agreement target is preferably 1.5 °C above pre-industrial levels. Roughly how much additional warming does "business as usual" represent compared to the preferred target?

a) About 0.5 °C more
b) About 1 °C more
c) About 2.5 °C more
d) About 10 °C more
Ans: ______

**41.** If a 1 $cm^3$ sample of a metal–organic framework has up to 94% empty (pore) space, as stated in the slides, approximately what volume of that sample is occupied by solid framework material?

a) 0.06 $cm^3$
b) 0.94 $cm^3$
c) 0.5 $cm^3$
d) 0.06% $cm^3$
Ans: ______

**42.** The slides state that increasing temperature always reduces surface coverage in physisorption, whereas in chemisorption, increasing temperature can favour adsorption. At very high temperatures, which type of adsorption would you expect to dominate, if both are otherwise possible?

a) Physisorption, since it is generally weaker
b) Chemisorption, since raising temperature can promote it while suppressing physisorption
c) Neither type occurs at high temperature
d) Both occur equally at all temperatures
Ans: ______

**43.** If a reaction has a theoretical yield of 50 g of product but only 35 g is actually recovered, what is the percentage yield, using the formula given in the slides?

a) 50%
b) 70%
c) 35%
d) 143%
Ans: ______

**44.** Given that steam methane reforming accounts for about 50% of global hydrogen production and coal gasification for about 20% (per the slides), roughly what fraction of global hydrogen production comes from other routes (e.g., electrolysis)?

a) About 5%
b) About 30%
c) About 50%
d) About 80%
Ans: ______

**45.** Stratospheric ozone forms when UV splits $O_2$ into radicals that react with more $O_2$; tropospheric ozone instead forms from NOx and VOCs reacting in sunlight. Which of the following would most directly increase tropospheric ("bad") ozone levels in a city?

a) Reduced UV radiation reaching the stratosphere
b) Increased vehicle traffic emitting more NOx and VOCs on a sunny day
c) Reduced industrial activity at night
d) Increased ocean evaporation
Ans: ______

**46.** CFCs are broken down by UV light in the stratosphere (releasing ozone-destroying chlorine), while HFCs are broken down by hydroxyl radicals in the troposphere and contain no chlorine. Why do HFCs not deplete stratospheric ozone even though, like CFCs, they are potent greenhouse gases?

a) HFCs never reach the atmosphere
b) HFCs are broken down in the troposphere before reaching the stratosphere, and contain no chlorine to begin with
c) HFCs absorb no infrared radiation
d) HFCs immediately convert to $CO_2$ upon release
Ans: ______

**47.** Second-generation bioethanol (from lignocellulosic waste) is described as carbon-neutral except for transport/biorefinery emissions, while first-generation biofuels (from food crops) carry their full lifecycle emissions. Why might second-generation biofuels be considered more environmentally favourable?

a) They require no biomass at all
b) They use waste biomass rather than food crops, largely avoiding the additional emissions of dedicated crop cultivation
c) They produce more $CO_2$ overall
d) They cannot be blended with fossil fuels
Ans: ______

**48.** Both the Haber–Bosch process and Ziegler–Natta polymerisation are highlighted in the slides as catalytic processes recognised by multiple Nobel Prizes. Based on the catalysts given for each, which pairing is correct?

a) Haber–Bosch uses Fe catalyst; Ziegler–Natta uses $TiCl_4$ with triethylaluminium
b) Haber–Bosch uses Pt catalyst; Ziegler–Natta uses Fe catalyst
c) Both processes use the same Ni catalyst
d) Haber–Bosch uses $ZnEt_2$; Ziegler–Natta uses Pt
Ans: ______

**49.** A hospital currently incinerates outdated, unused medication as its only disposal method. Applying the PPA hierarchy (prevent/reduce → recycle → treat → dispose), what would be the most preferable step to consider before resorting to incineration?

a) Continue incineration since it already counts as treatment
b) Investigate whether over-ordering of medication can be reduced at the source
c) Switch directly to landfill disposal instead
d) Increase the incineration temperature
Ans: ______

**50.** CFCs and $CO_2$ are both discussed in the slides in connection with atmospheric issues, but through different primary mechanisms. Which statement correctly distinguishes their primary environmental concerns?

a) CFCs mainly deplete stratospheric ozone; $CO_2$ mainly contributes to the greenhouse effect/global warming
b) CFCs mainly cause global warming; $CO_2$ mainly depletes the ozone layer
c) Both CFCs and $CO_2$ exclusively affect the ozone layer, not climate
d) Neither CFCs nor $CO_2$ have any documented atmospheric effects
Ans: ______

# Answer Key

| | | | |
|---|---|---|---|
| 1. | **c** | 26. | **b** |
| 2. | **b** | 27. | **b** |
| 3. | **b** | 28. | **c** |
| 4. | **c** | 29. | **b** |
| 5. | **a** | 30. | **a** |
| 6. | **c** | 31. | **c** |
| 7. | **c** | 32. | **b** |
| 8. | **a** | 33. | **b** |
| 9. | **c** | 34. | **b** |
| 10. | **c** | 35. | **b** |
| 11. | **b** | 36. | **b** |
| 12. | **b** | 37. | **b** |
| 13. | **c** | 38. | **a** |
| 14. | **d** | 39. | **a** |
| 15. | **b** | 40. | **c** |
| 16. | **b** | 41. | **a** |
| 17. | **b** | 42. | **b** |
| 18. | **b** | 43. | **b** |
| 19. | **b** | 44. | **b** |
| 20. | **d** | 45. | **b** |
| 21. | **b** | 46. | **b** |
| 22. | **c** | 47. | **b** |
| 23. | **b** | 48. | **a** |
| 24. | **c** | 49. | **b** |
| 25. | **b** | 50. | **a** |